

Q1

```
Last login: Fri Apr  3 16:23:14 on ttys000
asjadraza@Asjads-MacBook-Air-2 ~ % cd Desktop/OS
asjadraza@Asjads-MacBook-Air-2 OS % gcc -o gcc -o Q1 Q1.c -lpthread && ./Q1
[No Mutex] Final balance = 999 (expected 0, likely incorrect)
[With Mutex] Final balance = 0 (expected 0)
asjadraza@Asjads-MacBook-Air-2 OS %
```

Q2

```
asjadraza@Asjads-MacBook-Air-2 OS % gcc -o gcc -o Q2 Q2.c -lpthread && ./Q2
Thread 1 printing: 1
Thread 1 printing: 2
Thread 1 printing: 3
Thread 2 printing: 1
Thread 2 printing: 2
Thread 2 printing: 3
Thread 3 printing: 1
Thread 3 printing: 2
Thread 3 printing: 3
asjadraza@Asjads-MacBook-Air-2 OS %
```

Q3

```
asjadraza@Asjads-MacBook-Air-2 OS % gcc -o gcc -o Q3 Q3.c -lpthread && ./Q3
Student 1 sits on computer
Student 2 sits on computer
Student 3 sits on computer
Student 2 leaves computer
Student 4 sits on computer
Student 1 leaves computer
Student 3 leaves computer
Student 5 sits on computer
Student 6 sits on computer
Student 4 leaves computer
Student 7 sits on computer
Student 5 leaves computer
Student 6 leaves computer
Student 7 leaves computer
asjadraza@Asjads-MacBook-Air-2 OS %
```

Q4

```
asjadraza@Asjads-MacBook-Air-2 OS % gcc -o gcc -o Q4 Q4.c -lpthread && ./Q4
Data Produced
Data Consumed
Data Produced
Data Consumed
Data Produced
Data Consumed
Data Produced
Data Consumed
Data Produced
Data Consumed
asjadraza@Asjads-MacBook-Air-2 OS %
```

Q5

```
asjadraza@Asjads-MacBook-Air-2 OS % gcc -o gcc -o Q5 Q5.c -lpthread && ./Q5
I am in 1
I am in 2
I am in 3
I am in 4
I am in 5
I am out 1
I am out 2
I am out 3
I am out 4
I am out 5
I am in 1
I am in 2
I am in 3
I am in 4
I am in 5
I am out 1
I am out 2
I am out 3
I am out 4
I am out 5
I am in 1
I am in 2
I am in 3
I am in 4
I am in 5
I am out 1
I am out 2
I am out 3
I am out 4
I am out 5
I am in 1
I am in 2
I am in 3
I am in 4
I am in 5
I am out 1
I am out 2
I am out 3
I am out 4
I am out 5
```